OUROBOROS KA-52M



Indexes of stable builds:
2VFO (A/B) + second bank (C/D).
3VFO (A/B/C) + second bank (D/E/F).
In the version with FM radio, Ka-Viewer is removed.
The h index is a large FM radio skin.
Index d - small skin of the FM radio.
Index w - by F0, the list is called
WFM broadcast channels or all of them channels. Added the game "Mancala" (beans from Nokia) byF+Fn.

Version without index V7***
Working (experimental) version,
which will be developed further.
Without: viewer, scrambler, F0 radio, games.
For those who just love flashing and testing!

COMPLETE USER GUIDE
FIRMWARE UPDATES

Quansheng UV-K5V3 и UV-K1
(Bootloader 7.★★.★★)

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Firmware Features

- Version with double and triple VFO.
- Single VFO/MR mode.
- Assignable side PTTs.
- Assignable side buttons and menu button (short and long press).
- Reception in FM/AM/DSB/CW and WFM modulation (broadcast stations). Transmission: FM / DSB/CW (if the complete ban on transmission is lifted in F Lock, PTT works in AM mode).
- Expanded number of bandwidths, "inflated" to 32, for working with satellites.
- Enable Doppler correction for double VFO.
- Functional shortcut menus with hot keys for controlling sub-tones and repeater shift, LNA/AGC/RF GAIN settings from VFO or spectrum mode.
- Scan list system (up to 20 lists + Monitor). Arbitrary selection of multiple simultaneous scans (VFO or spectrum scans).
- Audio profiles (equalizer). Select an audio profile for FM and AM modulation.
- Extended microphone gain range.
- Day and night mode screens.
- Full-featured spectrum analyzer with four scanning modes (S LIST, BAND, FREQ, RANGE).
- Scan through lists, by set ranges (bands), by borders, or around a frequency.
- Saving the scan history with editing. Quick access to scan history, save frequencies from history to memory. Uploading your history to an app on your phone or computer.
- Enables a transmission ban on a prohibited frequency or any particular channel.
- FM radio. 36 fixed settings. Ability to record radio stations to channels with WFM modulation.
- Quickly save frequency, delete channels, rename channels.
- Turn on the constant illumination of the screen and flashlight in various modes with quick buttons.
- Duplicating the transmission of Morse characters with a flashlight.
- Flash of the flashlight when incoming RX.
- Use the buttons on the main MR/VFO screen to set frequently used parameters.
- Repeater (FM or CW auto beacon). The answer is "Roger" or Morse code.
- Battery voltage calibration: 1400 mAh, 1600 mAh, 2200 mAh, 2500 mAh, 3500 mAh..
- Remote control of the walkie-talkie from your phone or computer.

And much more...

1. It is important

You have purchased a new transceiver (walkie-talkie). Before installing any firmware, be sure to save the factory calibrations!

Why this is important... Calibrations in the transceiver are necessary to ensure maximum accuracy,

stability and quality of operation of the transceiver path. Since electronic components have manufacturing errors, the product is adjusted by instruments and software calibrations are made to it. You get a transceiver calibrated for your instance. Calibrations provide settings such as: SQL-noise reduction, RSSI-received signal level, PWR-transmitter output power, VOX-voice control level, BAT-battery setting, U/% - battery voltage and capacity readings, microphone level, master oscillator frequency, and others.

Recommendations for the adapter cord

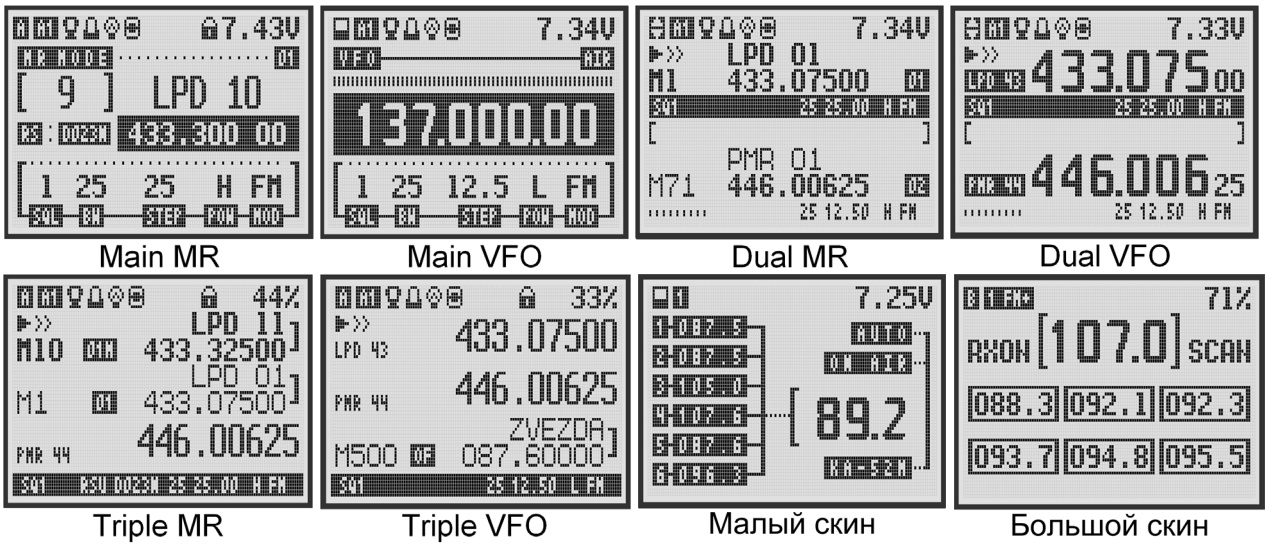
- Type–C cables are best suited for downloading firmware due to their speed. The cords supplied with the walkie-talkie are only suitable for charging (despite the note in the instructions: "use charging via the Type–C port only in emergency cases"), and the program does not see them.
- There are two types of adapter cords:CH340 and Prolific. The main difference lies in their stability, price, and driver issues. On the CH340, drivers are installed very simply and work stably. Prolific early versions suffer from problems with fake chips. Choose what best suits your equipment.
- To download the firmware, always switch the walkie-talkie to download mode first before plugging the CH340 cord into the walkie-talkie. The screen will be turned off and the LED will light up. Otherwise, the LED may not light up, information may be displayed on the screen, or dots may appear. This happens when one end of the cord is already plugged into your smartphone or PC, and the other is connected to the walkie-talkie, and you switch it to firmware mode.

Attention! Never pull the cord out of the device while downloading the firmware (LED flashes), trying to stop the process. This will inevitably lead to a serious malfunction. It will be impossible to restore the radio without "surgical" intervention.

- Do not switch on the radio station for transmission without an antenna and do not use mismatched antennas for communication. You can overheat and burn the output transistors of the transmitter.
- Do not increase the output power of the transmitter programmatically without using power measurement devices (Pmax = 5 W) and without monitoring current consumption (Imax = 1.5 A) during transmission.
- Despite the fact that the firmware allows you to receive signals at frequencies in the range of 12,000-900,000 MHz, the output stages have hardware limitations to two transmission ranges:UHF (400-470 MHz) and VHF (136-148 MHz).

The output power of the transceiver on other frequencies will be low and becomes irrelevant for over-the-air operation.

2. LCD display



Explanation of symbols on buttons and abbreviations

1 – Simple tap.
1 – Long press.
The MR Channel.
VFO – Frequency.
F1 – Click after the F button..

Display status bar

Icon / parameter	On Function	On Assignment / action
Frequency shift	F+Fn1 F+Fn2	Doppler shift correction is enabled. Shift polarity reversal.
[A][B][C]	2	Channel selection.
[MI][NR][MX]	F4	Audio profiles inFM, AM. ParameterSetAud.
VOX	F5	Activation of voice transmission.
Flash	7	Flash of the flashlight when RX is incoming.
Backlight	9	Screen illumination.
Repeater	8	Auto Beacon: responds toRX by transmitting a tone; in CW mode, transmits SOS.
Lock	F (long)	Lock the keyboard.
Battery	Меню of the"BatTxt" Menu	Percentage of battery capacity or voltage.
[1][2][3][4][5][6]	0	FMRadio: Memory register.
[1-6FM+]	8	FMradio: Extended radio frequency range.

Middle row of the display

Icon / parameter	On Function On	Assignment / Action
MR/VFO	3	Channel / Frequency mode.
[01]–[20]	★Scan	The number of the scan sheet.
[01M]	F+M (SLIST+M)	M-channel in the monitor list. It can be scanned in any scanning mode.
AIR, LPD 01…	F3 (MR)	Channel name (save).

RSU: [D023M]	F+M (subtones)	A digital subtone is set to receive.
[433.075 00]	F2 (VFO)	Channel frequency (save).
[RXON]/[NORX]	★ (long)	FMRadio: ControlRX (incoming).
[SCAN]/[STEP]	★ (short)	FMRadio: auto / manual search for a radio station.

Bottom line of the display

Icon / parameter	On functions	Assignment/Action
[SQ1]– [SQ9]	F+<>	Switching SQL (Noise Canceller): each bank has its own value (F+<> SQ1...F+<>SQ9).
[BW]	4	Bandwidth of the receiving path.
[STEP]	5	Receive frequency step in VFO.
[POW]	6	Transmit power.
[MOD]	0	Receive modulation.
[104.5,]	1-6–6	FMradio: memory cells (frequency, MHz).

For all VFO A, B, C, the bottom line of the screen with the parameters general and dynamic!
Which of A, B, C is selected, for that VFO and parameters are shown!

Doppler shift correction

The essence of the Doppler effect is to change the frequency and wavelength of sound that is perceived by a listener who is passing by a sound, such as a train or car signal. When the sound source approaches, its tone becomes higher, the waves are compressed, and when the waves are removed, they stretch and the sound gradually becomes lower. Since the nature of sound and radio waves is similar, when communicating with satellites whose frequency "floats" on Earth at a speed of several kilometers per second, it is necessary to take into account the Doppler effect.

3. Physical layout of radio station buttons

Button	Action button
PTT	Transfer button. Hold to go live.
SIDE1 (upper side): Fn1Shrt-short press Fn1Long – long press	The function is assigned in the menu. By default-MONITOR (noise canceller).
SIDE2 (lower side): Fn2Shrt-short press Fn2Long – long press	The function is assigned in the menu. By default-MAIN ONLY or another assignable function (see menu F2Shrt).
MENU (M[A])	Enter the main menu. Confirmation of the selection. Long press-flashlight. While scanning – go to the menu STILL.
UP/DOWN	Navigation in the menu, change the channel in MR, frequency step in VFO.
★SCAN	Short press: start scanning channels, switch the scan list (if the scan is already underway). Long press: Launches the SCANNER (frequency search mode). F+★ - Start the CTCSS/DCS.
0-9 tone scanner (numeric keypad)	Output frequency to VFO, channel number to MR.During scanning – enter the two-digit number of the scan sheet. Long press – special functions.
F (functional)	Long press – lock the keyboard. A short press activates F + [digit] (the F icon on the screen).

EXIT	Exits the menu. Stop scanning.

Long press of buttons (~700 ms)

0 – Change the FM/AM/DSB/CW modulation.
1-Select the VFO bank (ABC > DEF). Each bank has independent noise reduction parameters.
2-Switch between channels A > B > C (MR) or frequencies (VFO) (pointer on the screen).
3-Change MR/VFO modes on the selected channel.
4-Change the bandwidth of the receiving path.
5-Change the STEP step in the VFO.
6-Change of power H/M/L..
7-Flashlight indication when an incoming call is received.
8-Repeater (FMauto beacon). The walkie-talkie responds to calls with a signal. In FM modulation, you will get a sub-tone value in response. Subtract 200 from it and you will get the level of reception on the radio beacon. Roger set OFF or PIU; it is selected by frequency and duration. InCWmodulation, the beacon will respondwith an SOS or your text entered in the F+M window..
9-Turn the screen backlight on / off /return to timer mode.
M^A – Turn on the flashlight. Display the A or AB/ABC channel on the screenABC (select the $M^{A\ function}$ in the menu).
★ - Scanner mode (capture unknown frequency and sub-tone from a nearby walkie-talkie).
/F-Lock. Button blocking.

F+ and PTT+button combinations

F1-Transfer the frequency from MR to VFO (VFOchange the range in VFO).
F2-Delete the channel inMR, save the frequency inVFO. M^A-confirmation.
F3-Rename the channel toMR.. M^A-confirm.
F4 – Gain mode settings menu (see table).
F5-Enable voice transmission: in the VOX menu (OFF, 1 – max...9 – min).
F6-Enable transmission ban: XX – ban.
F7-Spectrum analyzer in scan-list mode.
F8-Spectrum analyzer in band mode.
F9-Spectrum analyzer in frequency mode.
F0-Turn on the FM radio. The appearance of the radio depends on the firmware.
F★ - Scanner mode (sub-tone capture at the selected frequency).
F+Fn1-Enable Doppler correction for double VFO (V.2VFO only.).
TO WORK ON SATCOM, SET THE BAND 32. This automatically activates the AFC for ±10 kHz Doppler correction.
F+Fn2-Change the direction of Doppler correction for a double VFO (V.2VFO only).
F+Fn2 – For VFO version 2VFO and VFO version 3VFO with index W..
PTT+Fn2 – Broadcasting a 1750 Hz tone (V.2VFOonly).
F+Arrows - Switch SQL (Noise canceller). Each bank has its own value.
F+M – Sub-tone settings menu and SLIST+M option for assigning a priority channel.

Subtones: what they are used for and how they work.

Subtones are an auxiliary analog or digital signal that is added to the main voice message. When the radio receiver is set to the same sub-tone, the speaker opens, and the caller can only hear you if the code matches. It doesn't hear other people working on your frequency or interference on the air. If the receiver has sub-tones turned off, it can hear everyone: with both enabled and disabled subtones. Examples of sub-tones for RX/TX: analog: 67.0 Hz-254.1 Hz; digital: D023N–D754I, D023I–D754I..

4. FMRadio Station

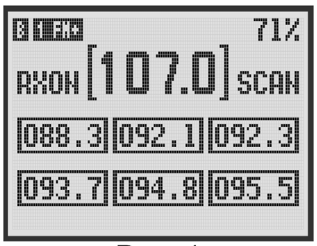


Рис. 1

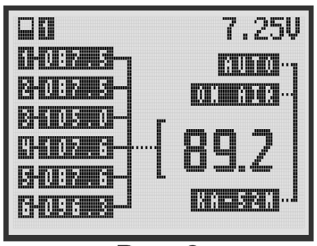


Рис. 2

The radio station receives radio broadcasts in the frequency range 76-108. 0 MHz. You can save up to 36 radio stations using the 6 fixed buttons and 6 memory registers (Fig. 1, 2). In order for the walkie-talkie to track the incoming signal from another walkie-talkie while listening to the radio, you need to set the parameter RXON (long press the ★ button) orNORX to disable tracking. The SCAN/STEP parameter is an automatic and manual station search, and can be switched by short pressing the ★button. A unique feature of the firmware is the recordingof FMradio stations in channels (see the section "Recording broadcasting stations in channels").

FM Radio button functions-радио

Button / command	Action
F0	Turn on the receiver.
★ - SCAN STEP	Auto Search. Manual search in frequency increments.
★ - RX ON NO RX	Incoming signal monitoring on the walkie-talkie. The incoming signal is not tracked.
UP/DOWN	Search for a radio station.
0	Change of banks (6 banks of 6 fixed frequencies = 36) .1
1-6	Six buttons for storing radio stations.
7	Save the radio station to a free channel.
8	Extending the FM band from 76.0-108 MHz.
M[A]	Turning on the flashlight.
EXIT	Exit

5. Transmitting Morse code

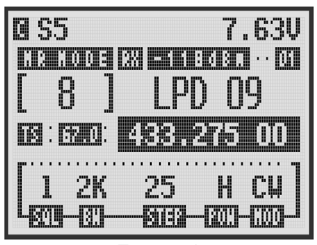


Рис. 1

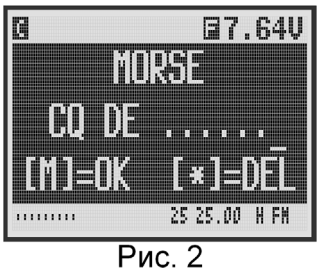


Рис. 2

To transmit Morse code, turn on CW modulation in the radio CW(Figure 1). The PTT tangent works as a key. You can receive on another radio in FM – then there will be a clear tone, or in CW - then 800 Hz is standard. To send a message from the keyboardCW, press F+MENU in CW modeMENU to enter the chat (Fig. 2). Text – maximum 15 characters. To quickly delete the last character, tap ★ Scan. Long press # to enter the blank "CQ DE", where CQ is an invitation for everyone to contact, DE is on behalf of the call sign. To transmit, press M. When the beacon CWis turned on, the repeater will respond in Morse code during CW (if you didn't enter anything in the chat, an SOS is sent). You can connect a wireless dongle via a customKenwood cord according to the diagram shown in Figure 3.

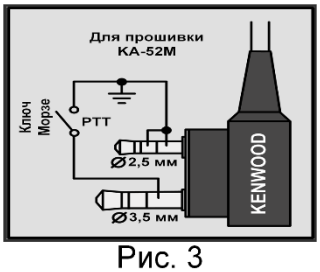


Рис. 3

6. Main Menu

№	Пункт Menu Item	Description
1	RxMode	Reception mode: SINGLE RX MODE – only one VFO. MULTI RX MODE-double/triple, TX switches to active. The hold time on the second VFO is set in RxHold. In RxHold, the OFF parameter restricts transmission on the secondary receiver from the primary PTT.
2	RxHold	In Dual Watch: Switches from the active channel to the passive channel on an incoming call. Parameters: OFF, 1-330s, LAST -stay on the last one.
3	Scramb	Enables the inverting scrambler in FM modulation mode. Inversion frequency-2600-3500 Hz, step – 100 Hz.
4	Roger	Enable Roger beep (end of transmission tone): OFF, OURO, KLAC, PIU, ICQ.
5	VOX	Voice transmission control-VOX. Parameters: OFF – disabled, 1-9-microphone sensitivity.
6	STE	Noise Canceller: Eliminates noise at the end of transmission. When the PTT is released, a short 55 Hz tone is issued to signal other transceivers to turn off the sound.
7	RP STE	Repeater" tail " noise reduction (Tail Elimination).
8	ScList	Select the scan list.
9	ScnTmr	Scan timer determines: Scan time inMR, VFO modes (STOP, 3s, 10s, 1min). Pause time after transmission (RXNOW, RX2s, RX5s).
10	ChDisp	Selects how channels are displayed on the display.
11	F1Shrt	Function assignment with a short side key press Fn1.1
12	F1Long	Function assignment with a long side key press Fn1.13
13	F2Shrt	Function assignment with a short side key press Fn2.14
14	F2Long	F2Long Function assignment with a long side key press Fn2.15
15	M Long	Function assignment with a long key press M^A..
16	M Shrt	Function assignment with by briefly pressing the M^A key.
17	KeyLck	Automatically locks the keyboard after a set time (0.15 s -10.0 min).F-selection of blocked elements: OFF KEY > KEY 015 s-10.0 min. Unlock-Long F (#) .18
18	TxOut	TxOut Transfer Timeout (TOT). The time that PTT can be held during transmission (in seconds). Protection against accidental pinching PTT.
19	BatSav	Power Saving (Battery Save): OFF/1:1/1:2/1:3/1:4/1:5.20.
20	Mic	Microphone sensitivity (+6.0.0 dB...+31.5 dB).
21	POnMsg	Enables / disables the display of a message when it is turned on (Power On Message).
22	BLTime	The time when the screen backlight is active when the buttons are pressed.
23	BLMin	Minimum illumination brightness.
24	BLMax	Maximum illumination brightness. If you hold down F in the active position , you can turn off the LED indicator for receiving and transmitting.
25	BLTxRx	Transmit/Receive backlight: OFF/TX/RX / TX+RX.
26	SetInv	Enables color inversion of the walkie-talkie screen.
27	BtnInv	K1 > K5: change the direction of the UP and DOWN buttons.
28	SetTmr	Setting the display of the RX/TX timers on the screen.
29	CWBL	Flashing of the flashlight in time with PTT in CW mode.

30	CWSPD	CWSPD Pause length of Morse characters for a double key (30-220 ms). Default 60 ms.
31	TONETX	Tone to air when pressing PTT+Fn2.F=1750 Hz default in increments of 10.
32	F	Lock Frequency lock setting. LOCK ALL – everything is forbidden. UNLOCK ALL – everything is allowed. UNLOCK ALL+WFM – plus radio broadcasting frequencies (use only for engineering purposes). Long F activation in UNLOCK ALL.
33	FrCali	Quartz Frequency Calibration BK4829.
34	BatTxt	4829.34 TXT Battery Charge Display Type: % > V > None.
35	BatCal	BatCal Battery Calibration.
36	BatTyp	Battery capacity selection: 1400 > 2500 > 1600 > 2200 > 3500 mA * h
. 37	Reset	Resets saved channels (NoCH) or completely resets settings, including saved channels (ALL).

6.1. Assignable functions for the side buttons Fn1, Fn2

Function	Action
NONE	No action.
FLASHLIGHT	Turns on / off the LEDlight.
POWER	Changes the power level.
MONITOR	Forced opening of the noise canceller.
SCAN	Starts scanning channels.
FM RADIO	In firmware versions with FMradio.
LOCK KEYPAD	Lock / unlock the keyboard.
VFO A/B/C	Switching A/ B..
VFO MEM	Mode change MR/VFO.
MODE	Modulation change.
RX MODE	Cycle режимов of reception modes (SINGLE RX MODE/MULTI RX MODE).
PTT	Функция The PTT function for an inactive channel.
WIDE / NARROW	Changes the receiving bandwidth.
RX AUDIO	Selects the audio correction profile.
RPT REV	Reverse shift for the channel (icon with two arrows).
SQL+	Increase the noise reduction level.
SQL–	Lowers the noise reduction level.

6.2. Hotkeys for MR/VFO modes

Function	Button	Value
STEP	5	Frequency step in VFO.
Power	6	Transceiver transmitter Power: H, M, L..
Mode	0	Demodulation mode: FM, AM, DSB, CW.
SAVE CH	F2	Save frequency to memory channel in VFO.
DEL CH	F2	Delete memory channel in MR..
REN CH	F3	Change channel Name memory in MR..
VOX	F5	Activation of voice transmission.
FLOCK	F6	Enable transmission ban: XX-ban.
TX CTCSS, DCS DX CTCSS, DCS	F+M	Sub-tone settings menu and option to assign a priority channel.

SLIST+M		
SetNFM (BW)	4	Filter bandwidth.
LED flash in RX mode	7	Light indication of an incoming call.
LCD Screen Backlight	9	Enable or disable permanent illumination.
SQL	F+Arrows	Switch the noise canceller. Each bank has its own value.

6.3. F+M Fast window of subtones, shift, S-LIST+M, CWand Morse code CW.

Parameter	Assignment
S-LIST+M	Flag of the monitor (priority) list.
RxDCS/RxCTCSS	Receive Subtone (DCS / CTCSS).
TxDCS/TxCTCSS	Subtone per transmission (DCS / CTCSS).
OFFSET	Frequency shift TX (MHz).
TxDIR	Shift sign (+/−).
Navigation	2/8 — item; 4/6-value; applies immediately.
In CW mode	, Instead of sub — tones, a Morse code text window; MA — transmission.

6.4. F4 – Fast gain window/AGC / audio profile

Parameter	Value Parameter	Appointment
ATT Assignment	0 > – 19 > – 16 > – 11.	An attenuator.
AGC	AUTO > MAN > FAST > NOR > SLOW > OFF.	Auto / Manual AGC mode.
GAIN	AUTO	Total path gain.
LNA	0–7 (MAN).	LNA amplification (manual mode).
DMPF	ON > OFF.	Demphasis.
SetAud (FM/AM)	MIN > NORM > MAX > OFF.	Audio profile (equalizer).
TXDev	is 0-25 (default is 12-13).	TX deviation: don't overstate it.

6.5. Audio profiles in FM and AM modes

Audio profiles are set in the F+4 (SetAud) window. Below is a description of each profile:

SetAud – FM

Function	The Character
MIN function	is Minimum gain / Neutral frequency response. For a quiet environment.
NORM	is the default balanced profile. Comfortable sound with moderate gain.
MAX	Maximum gain/more open sound. Without excessive aggressiveness.
OFF	frequency response by default.

SetAud – AM

Function	Character
MIN	Sharp. Narrow intermediate frequency filter with low gain. More selective, with better filtering of neighboring channels. On strong signals, the sound may be sharper or slightly distorted, but it remains clear.
NORM	Is Standard. As close as possible to the standard firmware.
MAX	is Open. Wider intermediate frequency filter with high gain. More open and pleasant

	on weak signals, but some signals may sound muffled, especially aviation ones.
OFF	frequency response by default.

7. Copying FREQ CTCSS/DCS from another walkie-talkie



Рис. 1



Рис. 2



Рис. 3



Рис. 4

★ - Determination of the unknown frequency of a walkie-talkie with a sub-tone (Fig. 1, 2).

F+★ - Sub-tone detection at the current frequency of 433.100 MHz (Figs. 3, 4).

M^{A}-Frequency conservation with a sub-tone.



Рис. 5

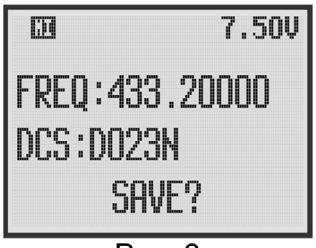


Рис. 6

Курсорами Use the UP and DOWN cursors to select a free channel, then press M^{A} again to confirm saving (Fig. 5, 6).

8. Easy scanning in channel mode (MR)

It is convenient to work with channels in the UV AIR TOOL editor on a PC, where for each channel the following parameters are written: channel name, frequency, bandwidth, frequency change step, power, modulation, sub-tone, scan list number and other parameters. The channel belonging to a specific scan list is specified in the "SL" parameter and has the following values:

1 ... 20-channel belongs to the scan list with this number.
NONE – the channel is not scanned.
MONITOR-channel in continuous scanning mode.

Button	Action
★Scan Action button	Starts scanning. Sequential selection of the scan sheet during scanning.
01...20	Select the required scan list during scanning.
00	Scan all scan sheets.
EXIT	Exit the scan.

All radio frequencies are distributed by user groups, for example: LPD, PMR, AIR, RIVER, MARINE, SATCOM, etc. In order for the walkie-talkie to scan a specific group or several groups selectively, you must assign the same scan list number to each channel in the group. For example: all channels in the LPD group are assigned scan list 01, channels in the PMR group are assigned scan list 02, and so on. Scanning in this mode is very fast, because the channel frequencies are known and there is no need to scan stepwise (the STEP parameter). Channels can be created and deleted by adding to your library of scan lists.

9. Simple Frequency Mode Scanning (VFO)

This scan mode is necessary to search for radio stations whose frequency is unknown. To start the search in frequency mode, briefly press the ★ Scan button. The frequency range will start scanning according to the set STEP. The STEP parameter affects the scanning speed: the smaller the step, the

lower the speed, but the probability of catching a radio station with a non-standard frequency is higher. Standard frequency means that groups of frequencies, depending on the band width and frequency, have different standard, generally accepted pitch between channels. But there are also frequencies in the intervals between the standard ones. Therefore, in bands, as will be described in the section "Scanning by bands", a step slightly smaller than the standard one is specially pre-set.
Use the UP and DOWN cursors to change the scanning direction. To stop and exit, click EXIT.

10. Scanning mode within specified limits (VFO)

Switch to multi-channel reception mode with the side button Fn2. Hold down button3 to switch to VFO frequency mode (Figure 1). At the top of the screen, enter the starting frequency (for example, 433.075 MHz). Click 2 to go to the bottom line and enter the end frequency (for example, 434.775 MHz). Go up again to the starting frequency. To start scanning, briefly press the ★ Scan button (Figure 2). Use the UP and DOWN cursors to change the scanning direction. To exit, press EXIT.



Рис. 1



Рис. 2

Simple scan management

Button	Action
★Action button Scan (briefly)	Run the scan inMR or VFO mode.
★ (during the scan)	Switch to the next available scan list. If the current one is empty, the device automatically moves to the next one.
01-20 (during the scan)	Direct input of the scan list number. Enter a two-digit number (for example," 05 " for scan sheet #5). The walkie-talkie will switch to him immediately.
00 (during the scan)	Switch toMonitor mode (scan all channels from 01 to 20).
M[A] (briefly)	Stop scanning on the current channel.
PTT (during the scan)	Stop scanning and start transmitting on the current channel.
UP and DOWN	Change the scanning direction.
EXIT (during the scan)	Stop scanning and go back to the last channel.

11. ★★★ Spectrum Analyzer ★★★

Use the following quick buttons to open different types of spectrum:

F7-S LIST (scan-lists). F8 – BAND (bands). F9 – FREQ (frequencies)
Button 6-switch modes (FREQ → S LIST → RANGE → BAND → FREQ).

The mode is restored when the walkie-talkie is turned on (if it was turned off directly in spectrum mode with the parameters saved by button 5).

Button	Spectrum Mode button	Action
F7	S LIST	Scan through channels from scan lists. UP/DOWN switches the scan list. Long press UP/DOWN to scan all channels.
F8	BAND	Scanning by bands. Button 4 – Band list selection menu. Use the UP and DOWN buttons to switch the ranges selected in the menu from the list. Total scan range in this mode: 24.890-462.000 MHz.
	FREQ	Scan in the range F (start)–F (stop). At startup, the current VFO is used as the center. When switching with the 6 button, the boundaries stored in EEPROM are used inside the spectrum.
F9	RANGE	Scan at the current frequency. UP and DOWN buttons – shift up or down from the current frequency.

11.1. Scan through Scan sheets (F7)

Press F7 to start the scan function (Figure 1). Press 4 to open the scan list selection menu. Use button 4 (check the boxes) to select the sheets that you want to include in the scan (Figure 2). Button 5 selects the current scan sheet, resetting the selection of the others. To exit and listen, click EXIT. You can use the UP and DOWN cursors to scroll through the scan sheets one at a time. To return to scanning all those selected, press and hold UP or DOWN.

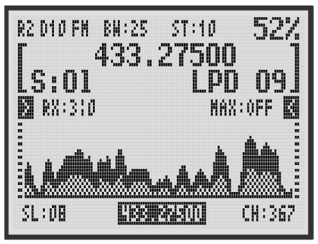


Рис. 1



Рис. 2

Fn2-add the frequency (interference) to the blacklist and continue.
1 - skip the frequency (interference) and resume scanning.

Creating a scan list

To create a scan list, follow these steps:
1. In the UV AIR TOOL editor, load channels from the walkie-talkie.
2. Create new channels and edit them.
3. Assign the channels a free scan list number inthe SL column.
4. Upload the channels back to the walkie-talkie.
5. Export the channels to your PC for use after a full reboot.

After the update, the name of the new scan list will appear in the scanner menu by pressing button 4.

11.2. Adding a channel to the priority scan list

This function allows you to create a priority list of channels for scanning, which is monitored independently of the current scan of scan sheets in any of the spectrum modes. On the screen, the letter M [01 M] is added to the icon with the scan list number [01M], which indicates that the channel is in the priority monitoring list. To activate this function, usethe F+M button combination to go to the sub-tone menu and set the S List+M parameter to ON. To enable scanning, go to theSPECTRUM menu (book 5) and set the Mon SL parameter to ON.

11.3. Scanning by band (F8)

Bands – frequency ranges with the initial and final scan frequency. For example: LPD = 433.075-438.800 MHz, PMR = 446.00625–446.19375 MHz, AIR = 118.0-135.9 MHz, and so on. Bands, like

channels, are loaded into the walkie-talkie using the UV AIR TOOL on your PC. They can be edited both in the program and directly in the walkie-talkie.

Press F8 to start the band scan function (Fig. 1). Use button 4 to open the menu (Fig. 2). Use button 4 to select the bands that you want to include in the scan. Button 5 highlights the current band, resetting the others. To listen, exit the menu with the EXIT button. When scanning with the UP and DOWN cursors, you can iterate through the selected bands in turn. Scanning across bands is slower than scanning across channels, because it runs with a set frequency STEP, but it is necessary to find an unknown station. All found frequencies will be recorded in the reception history.

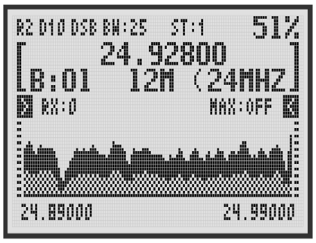


Рис. 1

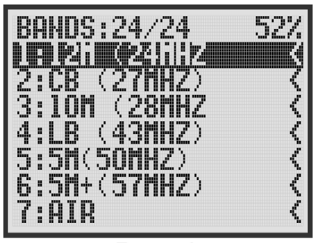


Рис. 2

Fn2-add the frequency (interference) to the blacklist and continue.

1 - skip the frequency (interference) and resume scanning.

11.4. Reception history

To enter the scan acceptance history, click 0 (zero). The # icon in front of the frequency indicates the frequencies that are blacklisted.

Button	Action button
2 and 8, UP / DOWN	Navigation up-and-and down.
Fn1	MON – open the noise canceller, FLOCK – lock on to the frequency, NORM – normal mode.
Fn2	# - the Blacklist tag. Allows you to remove / add a frequency to the blacklist.
1	Short press-SKIP (skip channel). Long press to show the channel name or subtone.
3	On / off history scanning.и.
4 (long)	Unloading history from memory.
5	Save the channel.
4 and 6	Sort history.
7	Save history to memory. The saved history can be uploaded to UV TOOL / UV DROID. If a subtone is defined in the history, it will also be uploaded to the app.
9 (long)	Clear your history.
0	Delete the channel from the history.

11.5. Table of uploaded bands

№	Название бэнда	Частота сканирования
1	12M (24 MHz)	24,89000–24,99000 MHz
2	CB (27 MHz)	26,51500–28,30500 MHz
3	10M (28 MHz)	28,00000–29,70000 MHz
4	LB (43 MHz)	43,30000–43,58750 MHz
5	5M (50 MHz)	50,00000–54,00000 MHz
6	5M (57 MHz)	57,00000–57,50000 MHz
7	AIR	118,00000–137,90000 MHz
8	136–144 MHz	136,00000–144,00000 MHz
9	2M (145 MHz)	144,00000–147,90000 MHz
10	SECURITY	148,00000–149,00000 MHz
11	RAIL	151,72500–156,00000 MHz
12	MARINE	156,00000–163,27500 MHz

13	VHF (162 MHz)	162,00000–170,00000 MHz
14	VHF (170 MHz)	170,00000–174,00000 MHz
15	SATCOM–L	240,00000–250,00000 MHz
16	SATCOM–H	250,50000–270,00000 MHz
17	RIVER	300,02500–300,51250 MHz
18	ARMY	300,51300–336,00000 MHz
19	RIVER-H	336,01250–340,00005 MHz
20	UHF (400 MHz)	400,00000–433,00000 MHz
21	LPD (433 MHz)	433,07500–434,80000 MHz
22	PMR (446 MHz)	446,00625–446,19375 MHz
23	450 (MHz)	447,00625–459,00000 MHz
24	460 (MHz)	460,00000–462,00000 MHz

11.6. Editing bands

To edit your bands, follow these steps:

1. Go to the band spectrum (F8).
2. Open the list of bands (4).
3. Open the Band editing window (6).

Кнопки управления Editing control buttons

Button	Action
2 and 8 button	Up/Down the list.
4 and 6	Select a row.
EXIT	EXIT Exit lines, save changes.
7	Saving bands.

11.7. Parameters in the SPECTRUM menu (book 5)

SPECTRUM — PARAMETERS (tab 1)

Button / command	Action
RSSI DELAY (2 ms)	RSSI measurement delay before frequency change (in ms) - affects the scanning speed.
SPECTRUM DELAY (0 s)	Time to hold the detected frequency before continuing scanning (0 = immediately on without delay).
MAX LISTEN TIME (OFF)	Maximum time to listen to a captured station before forcing scanning to continue (OFF = listen indefinitely while there is a signal).
NOISE LVL OFF (48)	Noise level to disable reception.
GLITCH MAX (15)	Threshold of" purity " of the signal based on the chip demodulation failure indicator.
NOISE HYSTER (13) The gap between the "open"	and "close" thresholds (the opening is always lower than the closing by this value) of the noise canceller.

RECORD TRIG (20)	The threshold at which auto-recording is triggered (if enabled).
SET AUD (A1 MIN)	The same as in STILL – quick selection of the audio profile.
MONITOR SL (OFF)	List of frequencies for continuous monitoring mode.

SPECTRUM-RANGE SCAN parameters (tab 2)

Button / command	Action
FSTART (xxx. xxx)	Initial frequency of the scan range (in MHz).
FSTOP (xxx.xxx)	The final frequency of the scan range (in MHz).
MODULATION (FM)	Тип Modulation type: FM / AM / DSB/SW.

SPECTRUM — SETTINGS (tab 3)

Button / command	Action
POWER SAVE (OFF)	Save energy between scan cycles (delay/sleep).
KEY UNLOCKED	Auto-locks keys.
PTT: (LAST RX)	Behavior when pressing PTT during spectrum display.
RX BACKLIGHT (ON)	Turning on the backlight when receiving a signal.
CLEAR HISTO ALL	Clears the entire history.
CLEAR HISTORY N BL	Clear history (without blacklist).
CLEAR HISTORY BL	Clearing the blacklist.
1/FN2 KEYS (SKIP/BL)	Assign buttons 1 and Fn2 in spectrum mode: skip or add to the blacklist.
SPEC VIEW (SHADOW)	Display style of the spectrum graph.
RESET DEFAULT	Completely resets the settings to factory settings.

11.8. Buttons in SPECTRUM mode (scan)

Button	Action
★ Action button and F	Raise/lower noise reduction capture threshold.
1	By default SKIP / BL – skip (setting in the "Spectrum" menu).
Fn2	Defaults to BL/SKIP (setting in the "Spectrum" menu).
2 and 8	Move the lists up or down.
2	Manual control of the backlight (in the BL TIME menu-by time).
3	Change the BW bandwidth BW(8.33 kHz, 12.5 kHz, 25 kHz...).
4	In S LIST: list of scan sheets, in BAND: list of bands, in RANGE: zoom...
5	Open spectrum parameters. Save changes to the parameters. In scan history: save the frequency to a free channel.
6	Next mode: FREQ → S LIST → RANGE → BAND.
8	Changing the view: CLASSIC → BIG.
9	Modulation (FM, AM, DSB, CW).
0	Open history. Delete the current channel from the history.
UP	BAND: the previous band. S LIST: the previous scan list. RANGE: Move the range down. Long press: Scan all.
DOWN	BAND/S LIST: next. RANGE: move up. Long press: Scan all.
MENU	Switch to STILL mode.
PTT	Exit, copy frequency to VFO.
Fn1	Monitoring mode: NORMAL → FREQ LOCK.

STILL (raw registers of the BK4829 chip)

This is a manual adjustment of the receiving path itself. You don't need to touch it for no reason: by default, everything is already configured correctly.

The point	that this	is Greater Than / Equal	to Less Than / equal
to LNAs	is a rough gain step at the input (an attenuator for very strong signals).	The entrance presses harder – it saves you from overloading at powerful stations nearby.	More amplification – weak signals are more audible, but strong ones can "clog".
LNA	is the main gain of the input stage.	Weak signals are louder, but there is more noise at an empty frequency.	Quieter, but cleaner in noisy areas.
PGA	Gain after the mixer (fine tuning of the path volume).	Louder overall.	Quieter overall.
300Hz Cut	Low-pass filter coefficient (the "color" of the sound itself).	The sound is "thinner" /drier.	The sound is "fuller" /bassier.
Deemphasis	On / off compensation for high tops.	ON – correct for normal FM voice.	OFF – better for AM and non-standard signals, the sound is brighter.
HPF300 En	On/off the 300 Hz cut-off filter itself (separate from the coefficient above).	ON-cuts low frequencies/hum.	OFF – Low frequencies pass as is.
LPF3K En	3 kHz cut-off filter On/off (high frequencies).	ON-cuts out hiss/high noise.	OFF – high frequencies pass as is.
SetAud	Ready audio profile with one button (MIN/NORM/MAX/LOUD / MAX+ / OFF).	--	–
XTAL Cal	Quartz Calibration – accuracy of the entire frequency grid of the walkie-talkie.	Shifts the frequency to one side.	It shifts to the other side.
MIX	Mixer gain.	More gain in the path.	Less gain.
AFC Range	How wide the automatic frequency adjustment (AFC) can "pull" the frequency.	Wider correction range-keeps out-of-tune stations better.	Already – less risk of "dragging" to the next station.
AFC Off	On/Off auto-tune the entire frequency.	OFF – the frequency is not adjusted automatically.	ON (0) – normal operation, adjusts itself.
AFC Speed	How quickly the AFC adjusts.	Responds faster.	Slower, but more stable.

Important: The 300Hz Cut / HPF300 En / LPF3K En filters are applied immediately "live", but they change themselves when changing the frequency and modulation. If you want them to be saved on write, SetAud must be OFF. Ready-made profiles will immediately overwrite the manual settings.

The five main parameters are detailed:

Parameter	What is	More =	Less =
Noisee LVL OFF (Default: 48)	Noise threshold at which reception is closed.	Closes only when there is very dirty noise: catches weak/noisy messages, but more often gives false positives.	It closes faster-more strictly, false positives are less common, but you can skip a weak station.
Noise Hyster (default: 15)	The gap between the "open" and "close" thresholds (the open is always lower than the close by this value).	Stable, without "rattling", but less sharp reaction at the signal boundary.	It responds faster, but there is a higher risk of" blinking " reception on the border signal.
GlitchMax (default: 20)	Threshold of" purity " of the signal based on the chip demodulation failure indicator.	It is more tolerant of dirty / weak signals: it catches more, but more false positives.	Stricter — only pure signals, less often false positives, but it can skip weak ones.
RSSI Delay (default: 2)	Pause (ms) after setting the frequency before measuring the signal strength.	More precisely, I measured at each step — scanning is noticeably slower.	Scanning is faster — the measurement may not be accurate at high speed.
DynSQL (default: 5, 50 = off)	Threshold for a jump in signal strength relative to the background (dynamic noise reduction).	Need a sharper jump: stricter, fewer false positives.	Responds to the smallest jump — more sensitive, but more false positives.

Recommended bundle for the task:

- DynSql – lower (we catch the weak and close it clearly when the RSSI returns to zero).
- Noisee LVL OFF – no higher than default (~49), do not raise it to 60, otherwise the noise floor will not be closed. But it will also open to a signal in which you can barely distinguish something. Not a fact.
- GlitchMax – lower (self-excitation/intermode filter, which give a high RSSI, but a "ragged" structure).
- RSSI Delay – 2-3 ms, raise only for unstable readings on a fast scan.

11.9. Monitoring in the spectrum mode STILL (kn. Menu)

Item	Action
LNAS Action (3)	LNA Gain (Low Noise Amplifier) – the signal gain stage before the mixer.
LNA (7)	Gain of the first LNA stage: the higher, the stronger the signal, but more interference.

PGA (7)	Programmable Fole Gain Amplifier (after LNA) – balance between sensitivity and distortion.
300Hz CUT (9)	A cutoff of the frequency response in the region of 300 Hz reduces the effect of "boxiness" and "noise".
DEEMP HASIS (On)	Together with 300 Hz Cut, it improves the sound on the K5V3 walkie-talkie.
HPF 300 EN (ON)	High-pass filter.
LPF 3K EN (ON)	Low pass filter.
SET AUD (A1 NIN)	Select audio presets for FM and AM.
XTAL CAL (-3)	Adjusting the native quartz frequency at the reception.
MIX (3)	Gain: the higher, the better the dynamic range, but more noise.
AFC RANGE (0) The	range of AFC operation.
AFC OFF (0)	Disabling AFC.
AFC SPEED (52)	Speed of AFC operation.

Use buttons 2 and 8 to navigate through the low – level register menu.
To change values, use the UP/DOWN buttons.
Save the frequency from the spectrum menu STILL-by pressing 5.

General principle of "speed vs accuracy of reception"

The same trade-off works almost everywhere:

- Stricter thresholds (less Noisee LVL, less GlitchMax, more DynSQL, more Hysteresis) = fewer false positives, but higher risk of missing a weak station
- Softer thresholds = catches more, but more "garbage" stops on noise
- RSSI Delay and Step are the only two parameters that directly affect the speed of the scan itself (not the sensitivity): RSSI Delay is larger = slower and more accurate measurement; Step is larger = runs the range faster, but rougher.

12. Save and edit channels in the walkie-talkie

You can save the frequency in the walkie-talkie in several ways: from the scan history, by entering the frequency from the keyboard, and via the UV AIR TOOL app (see the section "Channel Editor").

Managing in edit mode

Button	Action button
0-9	Enter characters, numbers, and letters
*	Deleting the entered
<>character	Navigate to line
M	Confirm saving: {M}=YES NEW – yes new. {M}=REWRITE – overwrite the busy time

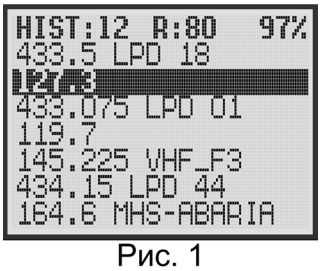


Рис. 1

12.1. Saving frequency from history

1. Enter the history by pressing the 0 button in scan mode (Figure 1).
2. Press button 5, the program will offer a free channel (YES NEW) (Fig. 2).
3. Press M to save, or first use the cursor (< >) to select another idle channel.

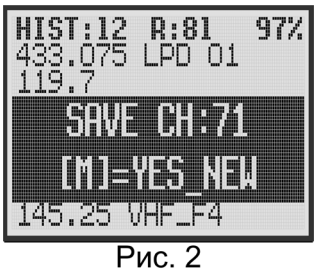


Рис. 2

12.2. Save frequency manually

1. In VFO mode, enter the frequency using the keyboard.
2. Press F2 - the program will offer a free channel (YES NEW) (Figure 3).
3. Press M to save, or first select cursors (< >) another idle channel.

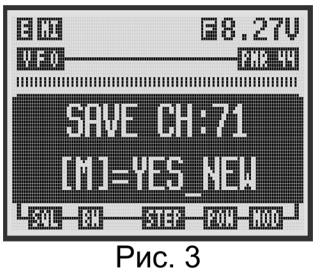


Рис. 3

If the last digits are not entered manually from the keyboard (the radio puts 0, 5 or resets the input instead of the desired digit), the main reason is an incorrectly configured frequency grid step (STEP). The walkie-talkie rounds the entered digits to the nearest step, which is currently set in the settings. Find STEP and choose a smaller step or universal 5.00 kHz / 2.5 kHz.

12.3. Editing the channel name

1. In MR mode, find the channel to rename in the walkie-talkie.
2. Press F2 to enter the edit window.
3. In the edit window, enter a name for the channel (Figure 4).
   Use the table "Manage in mode". editingoptions".
4. Save your changes - M..



Рис. 4

12.4. Editing channel parameters

To edit channels, use the following tables:

- Main menu.
- Hotkeys for MR/VFO modes.
- F+M Fast window of subtones, shift, S-LIST+M..
- Assignable functions for the side buttons Fn1, Fn2.

13. Special features

Function	Action
F+Fn1	Enables Doppler correction for double VFO (for V5.x).
F+Fn2	Change the direction of Doppler correction for double VFO (for V5.x).
PTT+Fn2	1750 Hz tone transmission (for V5.x).
Flash on RX	Flash on incoming signal (7).

X-BAND	Transmission forbidden: F+6.
F+M[A] (CW mode)	Opens the window for sending Morse characters. M[A] – sending.
F+M[A] (FM mode)	Window for sub-tones, frequency shift, and S-LIST/M (scan list and monitor list). Applied immediately.

Blocking the menu entry by pressing the "M[A]"button

In the "M Shrt" menu option, set any function, such as Flash Light.
Now, when you click the menu button, the flashlight will turn on. To disable it, press the M button, turn on the radio, and return the value to None in the parameter.

14. Battery calibration



Рис. 1

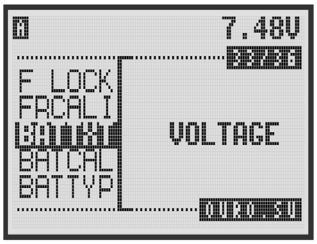


Рис. 2


Рис. 3

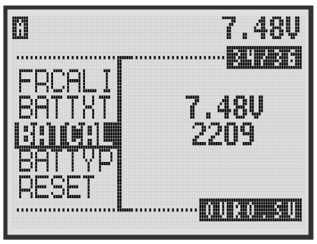


Рис. 4

In the Main Menu settings, set the following parameters:

1. BatTyp – battery capacity: 1400 mAh, 2500 mAh, 1600 mAh, 2200 mAh, 3500 mAh (Figure 1). The battery loses its capacity over time, so the set value in the BatTyp parameter BatTypwill display the percentage of remaining capacity inaccurately.

2. BatTxt (%or U) - display the percentage of battery capacity or voltage on the screen (Figure 2). The battery parameter that can be trusted is voltage; it does not depend on the BatTyp parameter, provided that it is properly calibrated.

3. BatCal (Battery Calibration) – калибровка battery calibration. It is necessary to measure the voltage on the back of the battery with a digital multimeter (Fig. 3) when the radio station is switched on and adjust the four-digit number so that the voltage readings on the multimeter and in the calibration settings coincide (Fig. 4). The state of charge of the battery during calibration does not matter. It is important to know the critical parameters of the battery: the maximum valueof Umax = 8.4 V and the minimum value of Umin = 6 V, and charge it in time. It is advisable to check the charger for excess voltage at the end of the charge, when the green LED lights up. It is convenient to measure with a multimeter at the battery terminals on the side connected to the walkie-talkie (Fig. 5).ATTENTION! DO NOT CLOSE THE TERMINALS! The voltage must not exceed the critical value, otherwise the memory may be faulty and need to be replaced. Overcharging above 8.4 V and deep discharge below 6V cause severe chemical stress, causing the battery to quickly lose capacity, wear out, or completely fail. It is not recommended to chargethe 1400 mAh and 1600 mAh batteries via the Type-C portlocated on the walkie-talkie case, much less turn it on and press gear while charging. In this walkie-talkie, it is designed mainly for downloading firmware. This does not applyto a 2500 mAh battery with a Type-C connector on the battery case.


Рис. 5

4. BatSav (SAVE) – power saving function. When the walkie-talkie is on but no one is talking, the receiver doesn't work all the time. Instead, it cycles to" sleep " for fractions of a second or a second, and then briefly turns on ("wakes up") to check for an incoming signal.

1:1-The receiver is active for 1 unit of time and "sleeps" for 1 unit of time. The battery is saved

moderately, and the walkie-talkie responds to an incoming call instantly, without delay.

1:2-The receiver is active for 1 unit of time and "sleeps" for 2 units of time. Energy savings are higher. During an incoming call, the first word of the other person may be slightly "choked" (the walkie-talkie needs time to wake up and open the noise canceller).

1: 4 is a very hard saving mode. The receiver is active for 1 unit of time and "sleeps" for 4 units of time. The battery lasts as long as possible, but there will be a noticeable delay (up to 1-2 seconds) during the call, which is why the beginning of the interlocutor's phrase is guaranteed to disappear. For most tasks, it is optimal to use 1:1 or 1:2 mode. 1:4 mode should only be turned on when it is critical to save the remaining battery power, and the caller knows that after pressing the transmit button (PTT), you need to pause for a couple of seconds before starting speech.

15. Game "Mancala" (beans fromNOKIA)

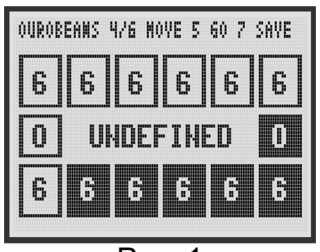


Рис. 1

The essence and purpose of the game: theice field consists of two rows of holes (6 pieces on each side for the player and the computer) and accumulative holes on the sides (Fig. 1).

Start: Bthere are 6 beans in each hole.

Goal:Collect as many beans as possible in your large storage well.

Rules of the game: you make a move by choosing any hole on your side, taking all the beans from it, which are automatically laid out in the next holes (including your own cumulative hole, but skipping the opponent's hole).

Extra Move: If your last bob falls into your own cumulative hole, you get the right to make another move.

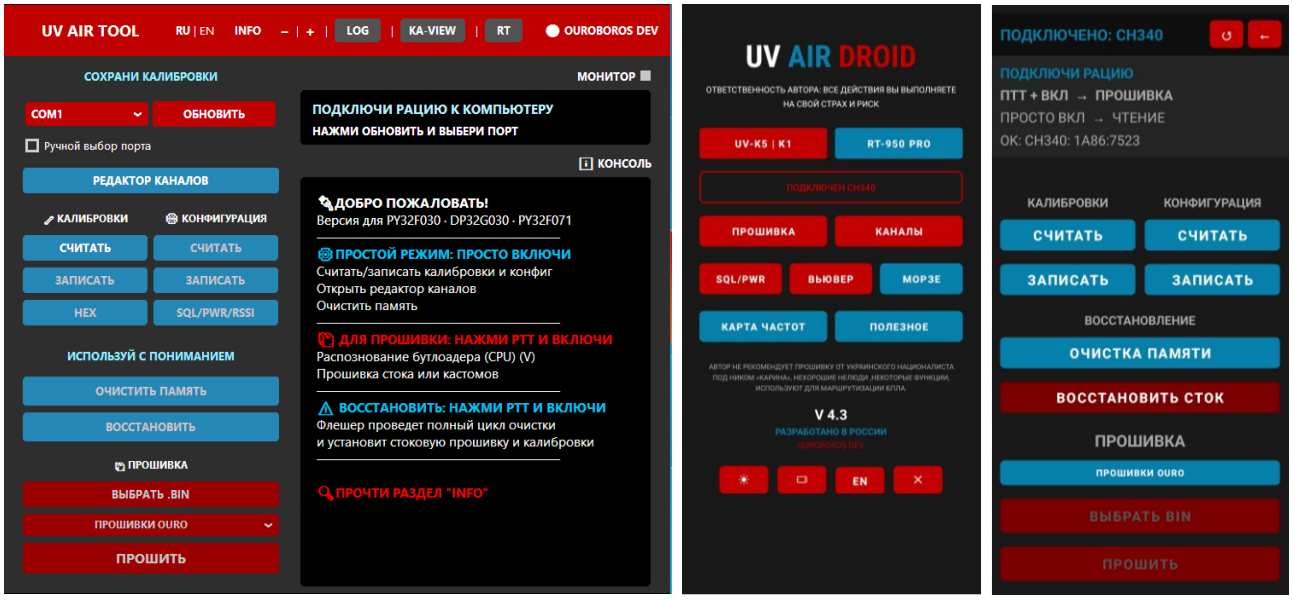
Capture of beans: if the last bean hits an empty hole on your side, and there are beans in the opposite hole of the opponent, you take both these opponent beans and your last bean to your cumulative hole.

End of game: The game ends when one of the players has no beans left on his side. The winner is the one who has collected the most beans in his hole in total.

F+Fn2-start the game, 4 <> 6-move through the holes, 5-pick up beans from the hole,
7-save the game (2VFO), 9 – reset the game (2VFO).

16. ПриложениеUV AIR TOOL and UV AIR DROID app

Application appearance



Use the UV AIR TOOL app on your PC and the UV AIR DROID app on your smartphone. Software the software allows you to perform the following functions:

- Read, write, and save factory calibrations and configuration of the transceiver.
- Uploading firmware to the transceiver from a file or the app's built-in storage.
- Edit, export, and import channel parameters.
- Clearingthe EEPROM memory.
- Crestoring the factory firmware after "fixing" the device.
- Edit noise canceller parameters:RSSI, NOISE, GLITCH.
- Setting the output power (PWR).
- Duplicate information from the walkie-talkie's LCD screen to a PC monitor or smartphone using the viewer.

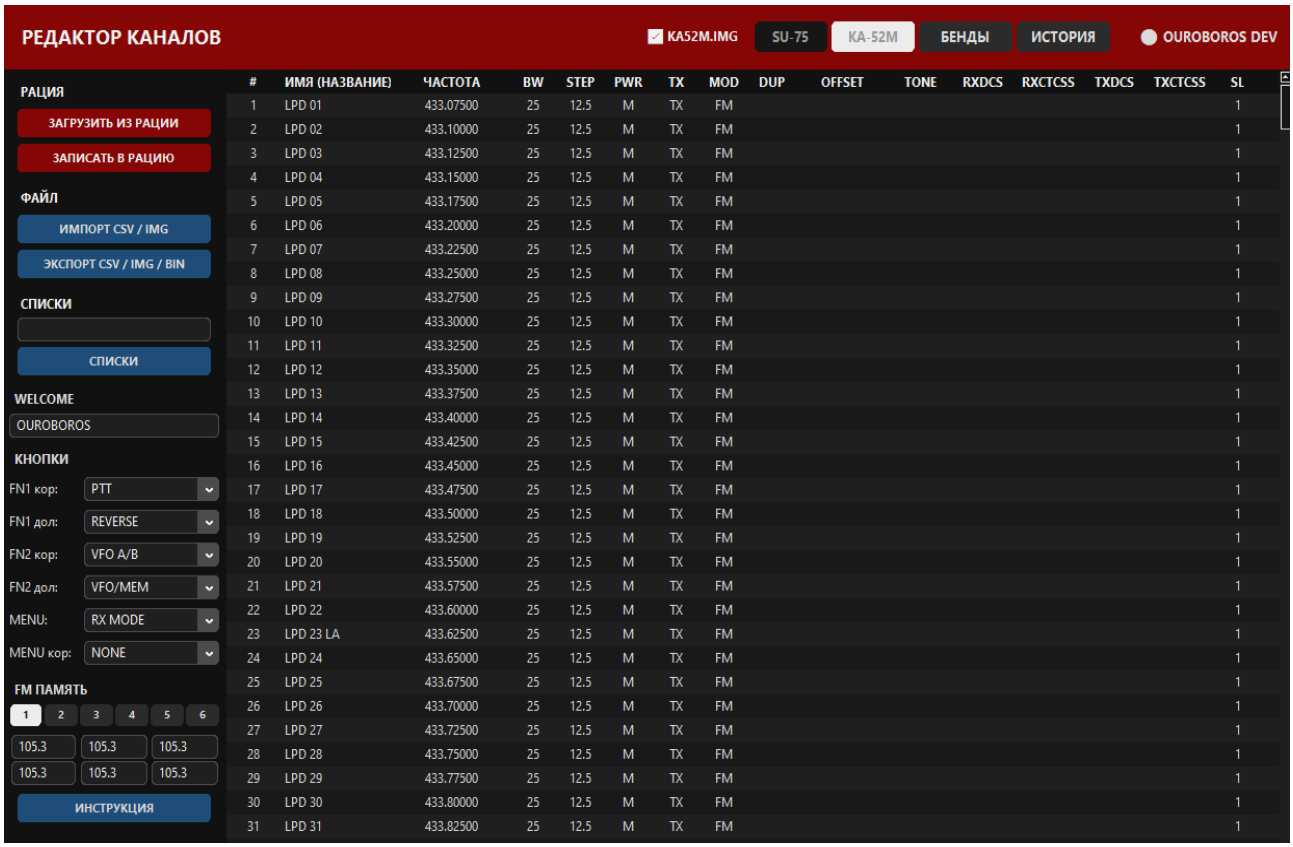
And other functions.

- The app is constantly updated.

Transceiver modes for firmware and data loading:

Status of the transceiver	Enabling	Action mode in the program
Simple mode	Turn on power	Reading and editing channels, calibrations, SQL, PWR. Clearing the EEPROM memory. Duplicate the LCD screen on the monitor and smartphone.
Firmware mode	While holding down the PTT transfer button, turn on the power - the LED will light up.	Downloading the firmware

17. Main menu of the app



17.1. Main Menu Options

Option	Function
RU | EN	Language selection.
INFO	Brief reminder.
+ | -	Adjusts the size of the app window.
LOG	Enables a log entry to be sent to the developer.
KA-VIEW	Viewer. Displays information from the LCD screen on the monitor.
RT	Working with RADTEL the RADTEL RT-950 PRO.
COM-X Walkie-talkie Manual	port selection.
UPDATE	Automatic detection of the radio's port and firmware.
CHANNEL EDITOR	Download, edit, import and export channels.
READ CALIBRATIONS	Read and save calibrations.
READ CONFIGURATION	Reads and saves the configuration.
RECORD CALIBRATIONS	Select a calibration file and upload it to the walkie-talkie.
RECORD CONFIGURATION	Select a configuration file and upload it to the walkie-talkie.
HEX	Editing of calibrations.
SQ | PW | S	Edit parameters:SQL, PWR, RSSI.
CLEAR MEMORY	Clears the EEPROM memory.
RESTORE	Restore factory firmware.
choose.BIN	Selects the firmware file to upload to the walkie-talkie.
FIRMWARE OURO	Selects the firmware in the app body.
FLASH	Starts the process of downloading firmware to the walkie-

	talkie.

17.2. Saving configuration and calibrations

To save your calibrations and configuration, do the following:

1. Enable simple mode.
2. Connect the walkie-talkie to the PC using an adapter cord.
3. Click "UPDATE". The mode, port, and firmware version will be determined.
4. Click "READCALIBRATIONS".
5. Specify the path to save the file-my_calibration. dat.
6. Click "READCONFIGURATION".
7. Specify the path to save the file-my_config. bin.

At the end, it is advisable to save the files as an archive in a safe place.

17.3. Installing the firmware

1. Turn on the walkie-talkie in firmware mode.
2. Connect the walkie-talkie to the PC using an adapter cord.
3. Click "UPDATE". The walkie-talkie type andBOOT are determined.
4. Tap "ПРОШИВКИ OURO FIRMWARE" to select the firmware from the app, or tap "SELECT .BIN" and specify the path to the firmware file.
5. Click "FLASH" and wait for it to finish.
6. Make a soft reset of the walkie – talkie settings (Reset-NoCH) when updating the firmware or a full reset (Reset – ALL) during the initial installation.
7. Set the parameters in the radio's menu items:SetInv, BtnInv and calibrate the battery (see "Battery calibration").

17.4. Recovery

The loss of calibrations after an unsuccessful firmware download leads to unmanageability of the station and often to "okirpichivanie" – a complete loss of performance. The factory firmware is protected from recording calibrations and configuration, which many people have questions about restoring. The app allows you to restore the walkie-talkie to its factory state automatically or manually.

To start the process, do the following:

1. Turn on the walkie-talkie in firmware mode.
2. Connect the walkie-talkie to the PC using an adapter cord.
3. Click "UPDATE". The walkie-talkie type and BOOT are determined.
4. Click "RESTORE".
5. Wait for the message: "Switch to BOOT mode".
6. Turn on the walkie-talkie in firmware mode.
7. Perform "CONTINUE RASKIRPICHIVANIE".
8. The factory firmware will load.
9. In the walkie-talkie menu, do "RESET ALL".

✓ Calibrations will be as close as possible to the parameters of the same type of transceiver.

17.5. Manual recovery

To start the process, follow these steps:

1. Turn on the walkie-talkie in firmware mode.
2. Connect the walkie-talkie to the PC using an adapter cord.
3. Click "UPDATE". The walkie-talkie type and BOOT are determined.
4. Download the firmware-ouro.calypso_v_v1.7.0.bin.
5. Turn on the radio normally.
6. Upload your configuration– my config.bin.
7. Upload your calibrations– my_c_calibration. dat.
8. Switch the walkie-talkie to firmware mode.
9. Download the stock (factory) firmware for:

   Quansheng UV–K1 – firmware.v7.03.01.bin

   Quansheng UV–K5V3 – firmware.v7.00.11.bin

✓ The calibrations will fully comply with the factory settings.

17.6. Clearing the EEPROM memory (if necessary)

Attention! Perform this if you have a file of native factory calibrations or, in extreme cases, calibrations from another walkie-talkie of the same type; otherwise, perform a restore, which automatically loads the appropriate calibrations and factory firmware. There are two ways to clear the memory:

Clearing the memory in the walkie-talkie

To clear the memory, do the following:

1. PressFand hold the F button to power up the walkie-talkie until the message "FULL WIPE ALL EEPROM HOLD F" EEPROMappears.
2. After starting the cleaning process (WIPING...) Releasethe F button..
3. When the message "CLEAN OK POWER OFF" appears, turn the power off and on again. "LOW BAT"appears. The EEPROM memory is completely cleared.

✓ Now you can download your calibrations and install stock firmware or any other software.

Clearing memory in the app

To start the process, do the following:

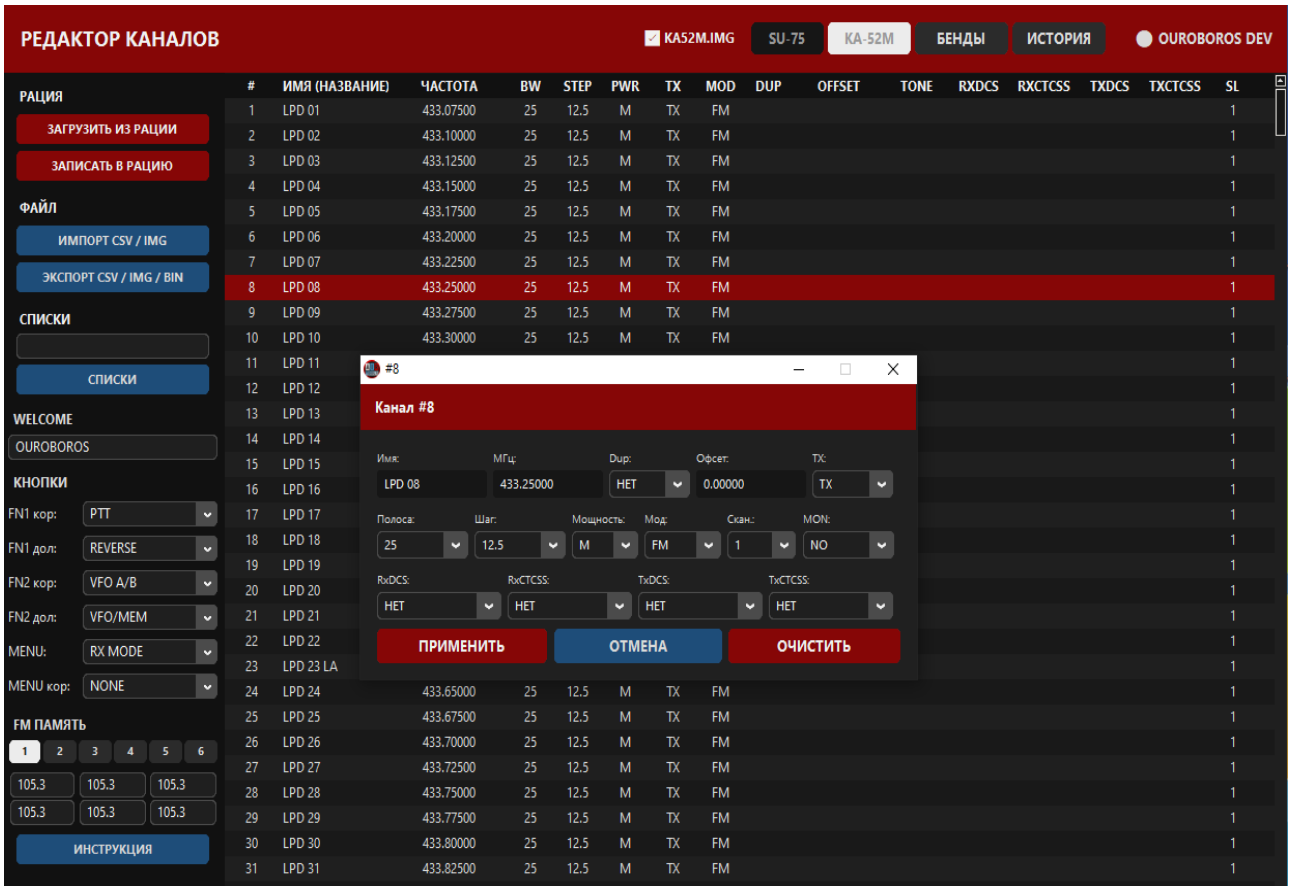
1. Turn on normal mode.
2. Connect the walkie-talkie to the PC using an adapter cord.
3. Run "CLEAR MEMORY".
4. Run "CONFIRM CLEANING!".
5. Wait for the cleaning procedure to complete
6. Click"RECORDCONFIGURATION".

   Specifythe location of the my_config filemy_config.bin and select it.
7. Click "RECORDCALIBRATIONS".

   Specify the locationmy_cof the my_calibration.dat file and select it.

✓ After finishing, you can download the factory firmware or any other one.

18. Channel Editor



Channels and bands are easiest to edit in the UV AIR TOOL app.

To log in to the editor, follow these steps:

1. Turn on simple mode on the walkie-talkie.
2. Connect the cable between the devices.
3. Tap "UPDATE"in the main menu. The walkie-talkie type and BOOT are determined.
4. Log in to the "CHANNEL EDITOR".
5. Use the context menu to start editing channels.

In the editor window, it will automatically load: LPD – 69 channels and PMR-16 channels. In Russia, they are used for free radio communication without registration or obtaining permits. There are a lot of operators working here, so it's best not to touch them. Other channels loaded by default may not be able to hear communication, as they may operate on different frequencies in different regions. You need to independently scan the air, find suitable frequencies in your area, and create your own channels and scan lists.

18.1. Channel Editor Options

Options	Function
KA-52M [SU-75]	Firmware selection for hardware UV-K5V3/K1 or [UV-K5V1].
BANDS	Display the spectrum band editor.
HISTORY	Uploads saved scanned channels from the walkie-talkie for editing.
UPLOAD FROM WALKIE-TALKIE	Upload channels from walkie-talkie for editing or exporting.

RECORD TO WALKIE-TALKIE	Uploading channels to the walkie-talkie after editing or importing.
IMPORT CSV / IMG	Upload channels from PC or smartphone to the editor in CSV/IMGformat.
EXPORT CSV/IMG/BIN	Save channels in CSV/IMG/BIN format on PC / smartphone.
LISTS	Sort channels by scan list list numbers.
WELCOME	Window for recording greetings on the LCD screen when the walkie-talkie is turned on.
BUTTONS	Assign functions to the side buttons Fn1, Fn2, and the MENU button.
FM MEMORY	6 memory registers for 6 fixed FMradio settings.
INSTRUCTIONS	Window for downloading firmware instructions.

18.2. Context menu of the Channel/Band/History Editor

Options	Function
The EDIT function	Calls the Edit parameters menu.
CHANGE THE SELECTED	PARAMETERS in the Group Change menu.
COPY Ctrl+ C	Copy channels/bands to the clipboard.
CUT Ctrl + X	Cut channels/bands.
PASTE Ctrl + V	Insert channels/bands from the clipboard.
INSERT WITH A DOWN SHIFT	Insert channels/bands with a top shift.
INSERT WITH UP SHIFT	Insert channels/bands with bottom shift.
DELETE WITH UPSHIFT	Delete channels/bands with bottom replacement.
CLEAR	the Deletion of channels / bands without replacing them.

18.3. Downloading channels from the walkie-talkie

1. Log in to the "CHANNEL EDITOR".
2. Click "DOWNLOAD FROM WALKIE-TALKIE".
3. If necessary, edit the channels using the context menu.
4. Save the channels to your PC by clicking "EXPORT CSV/IMG/BIN".
5. Upload channels back to the walkie-talkie by clickingRECORD TO WALKIE-TALKIE.

18.4. Uploading channels and bands to the walkie-talkie

1. Log in to the "CHANNEL EDITOR".
2. Upload your saved channels to the editor from your PC by clicking "IMPORT CSV/IMG".
3. If necessary, edit the channels using the context menu.
4. Tap "RECORD TO WALKIE-TALKIE". Bands will load automatically.

18.5. Editing bands

1. Log in to the "CHANNEL EDITOR".
2. Click on the "BANDS" tab.
3. If necessary, edit the bands using the context menu.
4. To save your changes, click "EXPORT CSV/IMG/BIN". The bands will be saved along with the channels.

18.6. Editing a story

1. Log in to the "CHANNEL EDITOR".

2. Click on the "HISTORY" tab. The Edit channels in history window opens.
3. If necessary, edit the channels using the context menu.
4. Copy or transfer channels from the clipboard to the channel editor.
5. Close the history window and paste channels into the channel editor.

18.7. Recording broadcasting stations in channels

#	ИМЯ (НАЗВАНИЕ)	ЧАСТОТА	BW	STEP	PWR	TX	MOD
500	ZVEZDA	87.60000	25	100	L	TX	WFM
501	XIT FM	88.30000	25	100	L	TX	WFM
502	SER DOZHD	88.80000	25	100	L	TX	WFM
503	DETSK RAD	89.20000	25	100	L	TX	WFM
504	RAD ENERGY	89.60000	25	100	L	TX	WFM
505	NOV RADIO	90.80000	25	100	L	TX	WFM
506	SPUTNIK	91.40000	25	100	L	TX	WFM
507	SHOKOLAD	91.90000	25	100	L	TX	WFM
508	KOMSOM PRV	92.30000	25	100	L	TX	WFM
509	STUDIO 21	92.70000	25	100	L	TX	WFM
510	DOROZHOE	94.20000	25	100	L	TX	WFM

Recording of broadcast channels

1. Prepare a list of broadcast radio stations.
2. Log in to the "CHANNEL EDITOR".
3. Fill in the "NAME" and "FREQUENCY" columns.
4. Select broadcast channels.
5. In the context menu, select "EDIT SELECTED".
6. In the edit window, in the "STEP" parameter, select the value 100.
7. In the edit window, in the "MODULATION" parameter, selectWFM.
8. Click "APPLY".

Control buttons in the FM radio channels window-радио

Button	Action button
F0	Open a window to listen to FMradio
2 and 8	Navigate
4	Name / Frequency
5	Transfer MR/VFO
6	Edit
EXIT	Exit

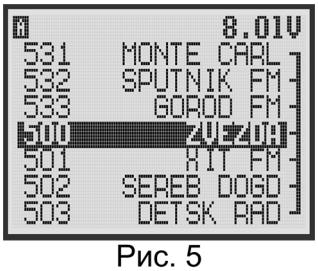
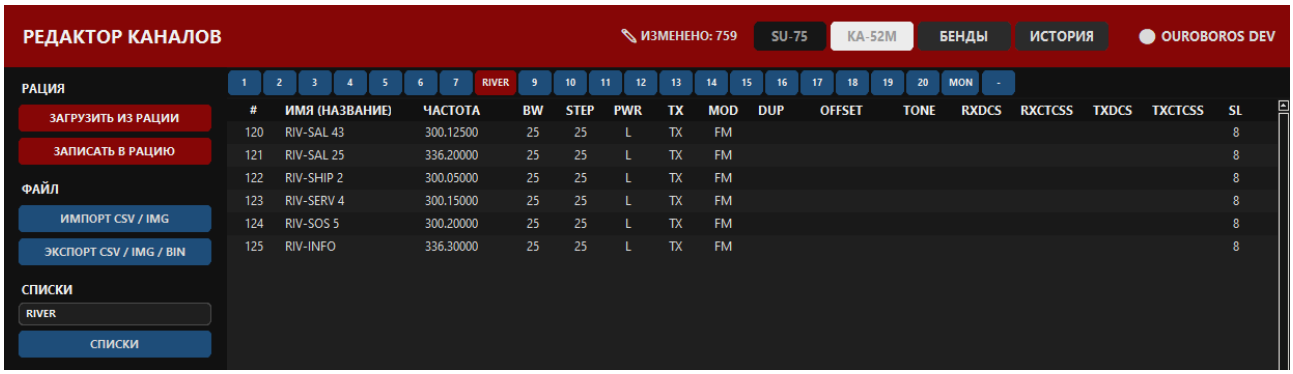


Рис. 5

18.8. Lists



The "LISTS" option allows you to view channels sorted by the SL parameter (scan list). You can edit, delete, and replace existing channels in this window, but you can't create new channels here.To do this, click "LISTS" again to return to the "CHANNEL EDITOR" window.

MAX Channel with firmware updates max.ru/channel_ouroboros.ru/channel_ouroboros

TG Channel with firmware updates t.me/filesouro

Yandex. Cart Chat t.me/fkFlbx_xnpU0M2Uy

RuTube rutube.ru/channel/31532419/videos

RuTube RUTUBE.ru/channel/31532419

Github github.com/igimalek

Cait ouro.su